

Advanced Database

Dr. M. Ali Nematbakhsh



Project4: B+ Tree

Question 1: Explain the Concept of B+ Tree and discuss why it is widely used for Database Indexing.

To answer this Question, I strongly recommend you watch this YouTube video and Write Down everything that you find Important for yourself: <https://youtu.be/aZjYr87r1b8?si=KAswNmYgoP0Ku94w>

Question 2: Based on the Suggested Video, or based on your knowledge, Explain Insertion and Deletion in the B+ Tree with a Simple Example (Keep it simple)

Question 3: Consider an initially empty B+ Tree. Insert the Sequence of keys: 2, 3, 7, 11, 17, 19, 23, 29, 31

For each of the following scenarios, draw the Tree structure:

A → The Tree has an order of 4

B → The Tree has an order of 6

Question 4: For each B+ Tree of Question 3, Show the Form of the Tree after each of the Following Series of Operations:

A → Insert 9

B → Insert 10

C → Insert 8

D → Delete 23

E → Delete 19

What to upload for us?

- Complete answers in one PDF file (It can be handwritten, etc)

All your files must be in one ZIP file named:

DBHW{Project Number}-{Student Complete name}-{Student Number}

Students must follow the Template structure, or they will receive a grade deduction.

Dead Line : 26 December - 5 Day

**Behnam Soufi
Ali Moeinian**